

Recent Progress in Utilization of Functionalized Organometallic Reagents in Cross Coupling Reactions and Nucleophilic Additions

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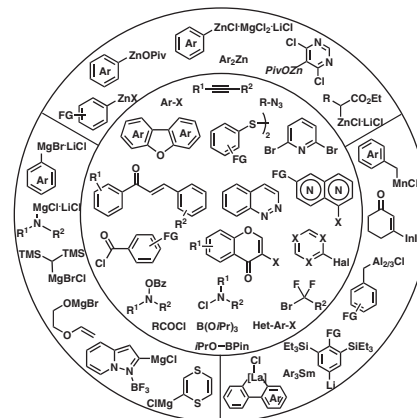
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Dedicated to Prof. Dr. Paul Knochel for his tremendous contribution to the development of novel organometallic reagents.



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Abstract Organometallic compounds have become increasingly important in organic synthesis because of their high chemoselectivity and excellent reactivity. Recently, a variety of organometallic reagents were found to facilitate transition-metal-catalyzed cross-coupling reactions and nucleophilic addition reactions. Here, we have summarized the latest progress in cross-coupling reactions and in nucleophilic addition reactions with functionalized organometallic reagents present to illustrate their application value. Due to the tremendous contribution made by the Knochel group towards the development of novel organometallic reagents, this review draws extensively from their work in this area in recent years.

Introduction

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Key words organometallic reagent, cross coupling, nucleophilic addition, transition-metal catalyst, synthesis

1 Introduction

Polyfunctional organometallics are one of the most useful reagents for the synthesis of various complex molecules and they play an important role in modern organic synthetic chemistry. As a result, many novel and efficient functionalized organometallic reagents, particularly, the Knochel



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type organometallic reagents (organometallic reagents containing LiCl as additive), have been successfully developed in the past decades.¹ These organometallic reagents are widely used in the field of agrochemical, pharmaceutical industry, cosmetics, dyes, natural products, and functional materials.² On the other hand, their extensive applications in synthesis were hampered by the moderate air- and moisture-stability of these organometallic reagents

(although recently, the Knochel group reported solid organozinc pivalates that are stable to heat, air, and moisture^{1h,9}). Related research on nucleophilic addition and cross-coupling reactions involving functionalized organometallic reagents has attracted increasing attention from chemists.

Transition-metal-catalyzed cross-coupling reactions between organometallic compounds and electrophiles, widely adopted in organic synthesis, are one of the most powerful tools for carbon–heteroatom (C–N, O, S) or carbon–carbon bond formation.³ For example, Heck, Negishi, and Suzuki couplings are great achievements in transition-metal-catalyzed cross-coupling reactions. Moreover, the nucleophilic addition reactions of organometallic compounds with unsaturated electrophilic reagents have also advanced in the past years.⁴ However, these reactions have some shortcomings. Some transition metal (such as palladium or nickel) catalysts often require expensive or non-commercially available ligands, and their industrial use is hindered by their high toxicity. Therefore, it is still desirable to develop more economical, efficient, and less toxic metal catalysts.

Recent years have seen the development of some inexpensive, efficient, and low-toxicity transition-metal catalysts, such as the cobalt salts, iron salts and manganese salt. These properties, combined with an excellent catalytic activity, have considerably improved the application of these catalysts in cross-coupling reactions with functionalized organometallic reagents. In addition, compared to traditional nucleophiles, the organometallic compounds offer the advantages of high reactivity, good chemoselectivity, and low toxicity.⁵ Thus, a number of extensions regarding the organometallic compounds have been developed. This review will focus on recent advances (from 2003 to 2020) in cross-coupling reactions and nucleophilic additions involving functionalized organometallic reagents (see Figure 1) to illustrate their practical applications in organic synthesis.

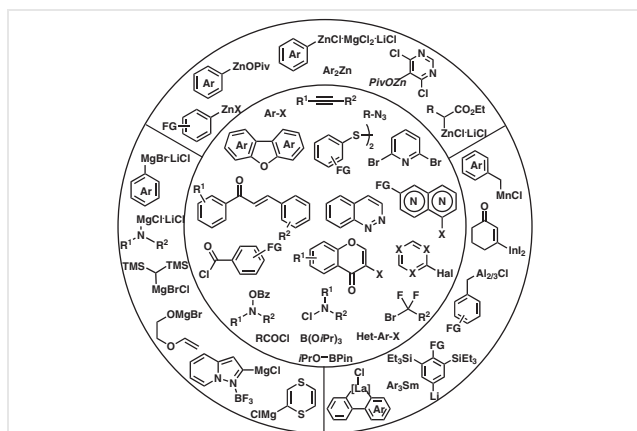


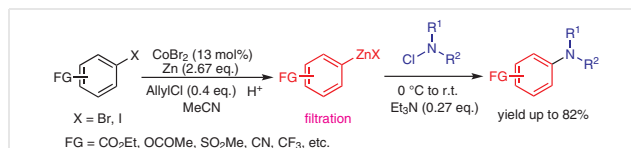
Figure 1 Recent developments of cross-coupling reactions and nucleophilic additions between organometallic reagents and various electrophiles

2 Transition-Metal-Catalyzed Cross Couplings Involving Organozinc Reagents

Organozinc reagents have been widely used in various processes. Compared to Grignard reagents, organozinc reagents enjoy better reactivity and unique functional group tolerance. Furthermore, the carbon–zinc bonds in organozinc compounds tend to undergo transmetalation with transition metals, which makes organozinc reagents valuable ingredients in cross-coupling reactions. Recent studies have shown that organozinc reagents partner well with substrates due to their decent functional group tolerance.

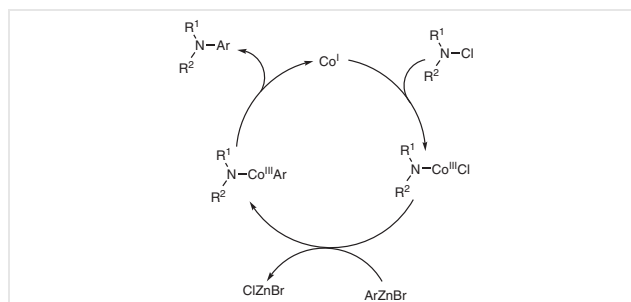
2.1 Cobalt-Catalyzed Cross Couplings Involving Organozinc Reagents

Cobalt-based catalysts, which are typically inexpensive and have low toxicity, have been considered most attractive; they can be used to achieve effective coupling of organozinc compounds and electrophilic reagents, and generally do not require sophisticated ligands for many transformations. Gosmini and co-workers⁶ reported a mild and highly efficient procedure for the amination of functionalized arylzinc reagents by secondary and tertiary *N*-chloramines using cobalt catalysis, and both secondary and tertiary arylamines were obtained in moderate to excellent yields (Scheme 1). This simple and convenient protocol displayed wide substrate scope and tolerated a large number of important functional groups.



Scheme 1

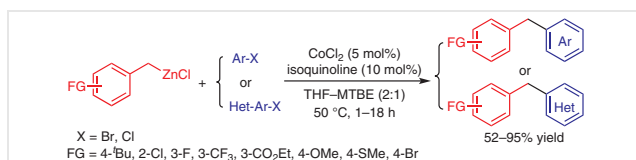
A plausible catalytic cycle is outlined in Scheme 2. The Co^I , which is formed in situ, catalyzes the formation of the arylzinc compounds in acetonitrile. The cobalt remains in the solution after filtration of the zinc dust, which allows further reactivity. First, the Co^I center undergoes an oxida-



Scheme 2 Possible mechanism of cobalt-catalyzed amination of arylzincs with *N*-chloramines

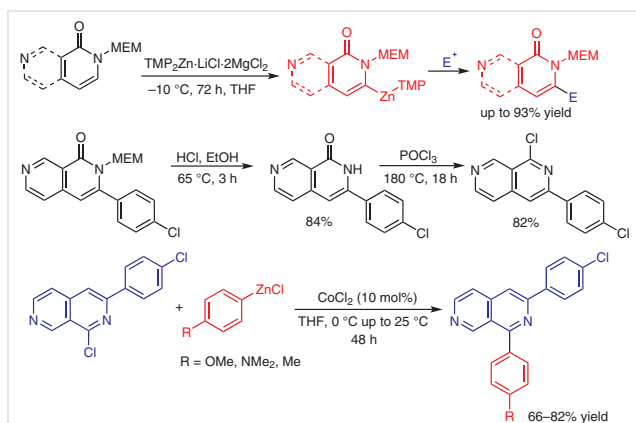
tive addition with the electrophilic *N*-chloroamine to afford Co^{III} species. This Co^{III} species is then transmetalated with the arylzinc halide to furnish $\text{R}^1\text{R}^2\text{N-Co}^{\text{III}}\text{-Ar}$. The cycle is completed by a reductive elimination, which produces the C–N product and regenerates Co^{I} .

Knochel and co-workers⁷ reported a novel and efficient cobalt-catalyzed and isoquinoline-promoted cross coupling between benzylic zinc reagents with various aryl and a series of heteroaryl bromides or chlorides, furnishing polyfunctionalized diaryl-, aryl- and heteroaryl-methane derivatives (Scheme 3). This procedure tolerates all kinds of functional groups (such as ketones, esters, nitriles), and proceeds under mild reaction conditions.



Scheme 3

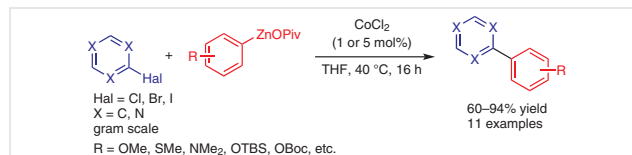
The same group⁸ developed a novel and general procedure for the regioselective metalation of MEM-protected 2-pyridones as well as MEM-protected 2,7-naphthyridone using $\text{TMP}_2\text{Zn}\cdot 2\text{MgCl}_2\cdot 2\text{LiCl}$ (MEM = methoxyethoxymethyl), which gave a series of new functionalized 2-pyridones and 2,7-naphthyridones. Subsequently, the functionalized MEM-protected 2,7-naphthyridones were treated accordingly and then furnished the bioactive 1-chloro-2,7-naphthyridines. Finally, the 1-chloro-2,7-naphthyridines with arylzinc halides underwent a cross-coupling reaction under the catalysis of cobalt(II) chloride (Scheme 4). This protocol gave the desired naphthyridines in satisfactory yields.



Scheme 4

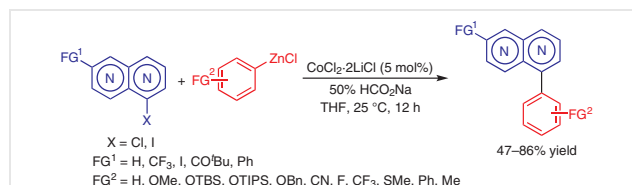
Hofmayer and Knochel et al.⁹ developed a practical and scalable cobalt(II) chloride catalyzed cross-coupling reaction of *N*-heterocyclic halides or halogenated aromatics with functionalized arylzinc pivalates under mild reaction conditions (THF as the solvent at 40 °C; Scheme 5). The sol-

id organozinc reagents PhZnOPiv (OPiv = OCOtBu) are stable to heat, air, and moisture for several hours in the air and for several years under argon. The advantages of this protocol include mild reaction conditions, a ligand-free system, the potential for large-scale reactions, and the low loading of catalyst (1–5 mol%).



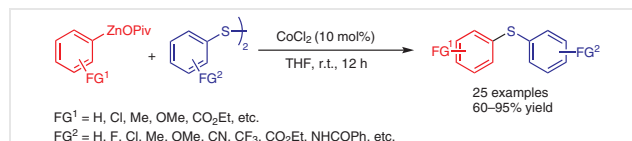
Scheme 5

Greiner and Knochel et al.¹⁰ reported an efficient synthesis of novel polyfunctionalized arylated naphthyridines by Co-catalyzed cross couplings between various halogenated naphthyridines and arylzinc halides, which offered the desired polyfunctional arylated naphthyridines in moderate to good yields (Scheme 6). The upsides of this protocol were that it was also suitable for the reaction of the aryl-magnesium with various halogenated naphthyridines. Mild reaction conditions, good yields and the broad substrate range make this protocol significant in organic synthesis.



Scheme 6

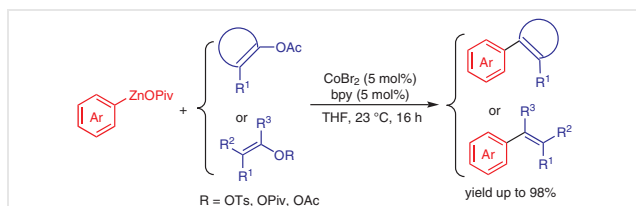
Dong and Knochel et al.¹¹ reported an efficient method for cobalt-catalyzed $\text{C}(\text{sp}^2)\text{-S}$ bond formation using solid organozinc pivalates and commercially available diaryl disulfides (Scheme 7). Various electron-rich and electron-poor organozinc pivalates were suitable substrates for the reaction protocol, and several sensitive functional groups including ester, nitrile, and amide were tolerated in the reaction. Room temperature conditions, good functional group tolerance and excellent yields were the advantages of this protocol.



Scheme 7

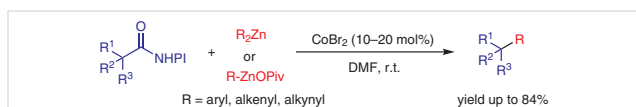
Li and Knochel¹² developed a practical electrophilic alkenylation reaction of various alkenyl acetates with functionalized aryl or alkenyl zinc pivalates under the systems of CoBr_2 (5 mol%) and 2,2'-bipyridyl (5 mol%) at ambient

temperature (Scheme 8). The excellent functional group tolerance and high stereoselectivity were well displayed in this protocol.



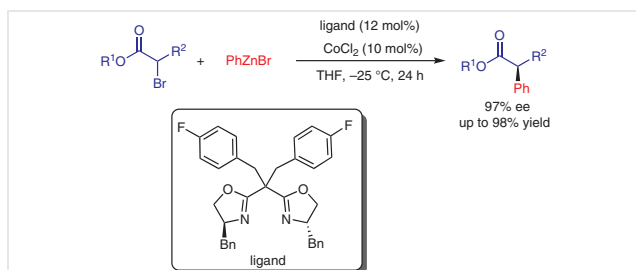
Scheme 8

Wang and co-workers¹³ reported an unprecedented cobalt-catalyzed decarboxylative Negishi coupling reaction of redox-active aliphatic esters with organozinc reagents in the presence of the solvent DMF at room temperature (Scheme 9). The reaction proceeded under mild reaction conditions with no additive or any external ligand, leading to diverse arylation, olefination, and alkynylation products with good efficiency.



Scheme 9

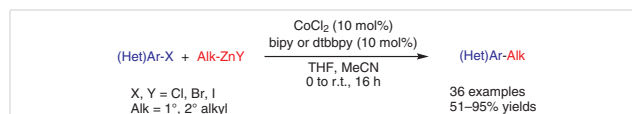
Liu and Bian et al.¹⁴ developed a novel cobalt-catalyzed enantioselective Negishi cross-coupling reaction that was achieved by arylation of α -halo esters with arylzinc halides. A variety of α -bromo esters were successfully coupled with various functionalized arylzinc reagents to provide enantioenriched α -aryllactones with high enantioselectivities and yields under mild conditions (Scheme 10). This protocol is suitable for gram-scale reactions and is of particular value to medicinal chemistry.



Scheme 10

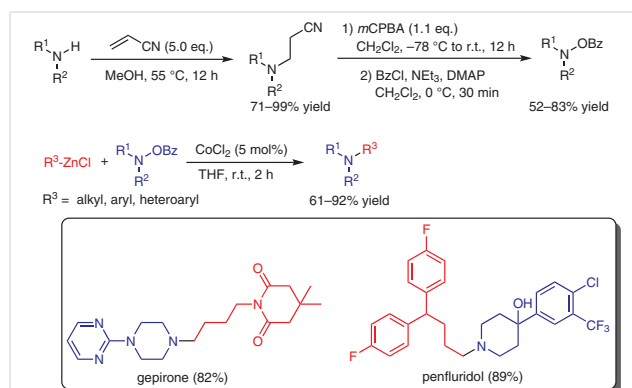
Lutter and Knochel et al.¹⁵ reported an efficient and practical cross coupling from various (hetero)aryl halides and functionalized primary as well as secondary alkylzinc reagents in the catalytic system of 10% CoCl₂ and 20% 2,2'-

bipyridine ligands (Scheme 11). In this protocol, the coupling with 1,3- and 1,4-substituted cycloalkylzinc reagents proceeded diastereoselectively to offer functionalized heterocycles with high diastereoselectivities of up to 98:2.



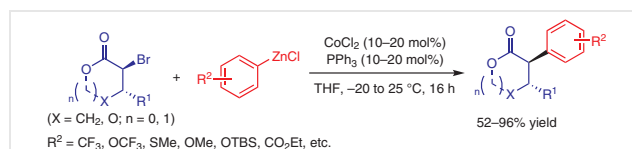
Scheme 11

Grassl and Knochel et al.¹⁶ reported an efficient access to polyfunctional hydroxylamine benzoates from the corresponding secondary amines; these polyfunctional hydroxylamine benzoates then underwent cobalt-catalyzed amination with various aryl-, alkyl-, and heteroaryl-zinc chlorides to give a range of polyfunctional tertiary amines in good yields (Scheme 12). It is worth noting that this method allowed the late-stage functionalization of various drugs, including peptidic target substrates, and can be readily used for the synthesis of complex target amines such as the pharmaceuticals gepirone and penfluridol.



Scheme 12

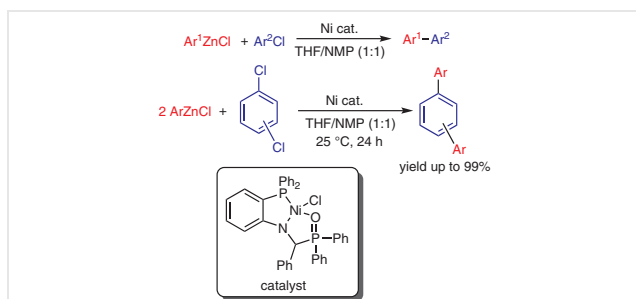
Hofmayer and Knochel et al.¹⁷ developed a highly *trans*-diastereoselective cobalt-catalyzed cross coupling of α -bromolactones with arylzinc chlorides, offering an efficient process that can be used to give optically enriched α -arylated lactones and protected aldol products (99% ee) in 52–96% yield in THF under mild conditions (25 °C, 16 h; Scheme 13). The methodology displayed a good substrate generality and excellent functional group compatibility.



Scheme 13

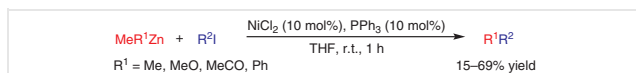
2.2 Nickel-Catalyzed Cross Couplings Involving Organozinc Reagents

Although the nickel-catalyzed coupling reaction of arylzinc reagents with electrophiles lacks some generality, sometimes it shows very high catalytic activity in some coupling reactions. Meanwhile, Ni-based catalysts could benefit a wide range of industrial processes. Wang et al.¹⁸ developed a new and efficient nickel-catalyzed Negishi reaction from arylzinc reagents and aryl chlorides in a mixed solvent (THF/NMP = 1:1) at room temperature (Scheme 14). This reaction exhibited low catalyst loading and good functional group tolerance.



Scheme 14

Pekel and Erdik¹⁹ reported a new, simple nickel-catalyzed protocol for the cross coupling of mixed methylarylzincs with primary alkyl iodides in THF in the presence of NiCl₂/PPh₃ at room temperature, which achieved the selective aryl transfer process and provided aryl-primary alkyl coupling products in moderate yields (Scheme 15). This protocol could be operated at room temperature within a short reaction time.



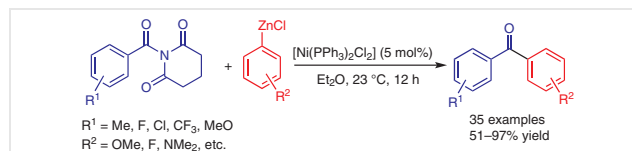
Scheme 15

Zhang et al.²⁰ reported an efficient Negishi cross-coupling reaction by using a Ni-catalytic system to prepare iso-flavones from 3-iodochromones or 3-bromochromones and arylzinc bromides in THF solvent at room temperature (Scheme 16). Notably, room temperature and inexpensive Ni catalysts would be economically beneficial to this protocol.



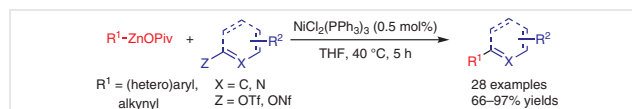
Scheme 16

Shi and Szostak²¹ developed the first Ni-catalyzed Negishi-type coupling of amides for the preparation of various polyfunctionalized diaryl ketones by chemoselective C–N bond activation from amides and arylzinc halides under exceedingly mild reaction conditions (Scheme 17). The advantages of this coupling reaction included mild reaction conditions, broad substrate scope and great functional group tolerance, and provided a new strategy for amide C–N bond/organozinc cross coupling.



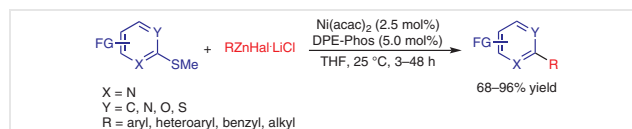
Scheme 17

Hofmayer and Knochel et al.²² developed an efficient and practical Ni-catalyzed cross coupling between various unsaturated nonaflates or triflates and (hetero)aryl or alkynylzinc pivalates, which gave the coupling products in high yields (Scheme 18). This protocol allowed a low catalyst loading (0.5 mol%) and obtained the coupling products in significantly high yields and good diastereomeric ratio.



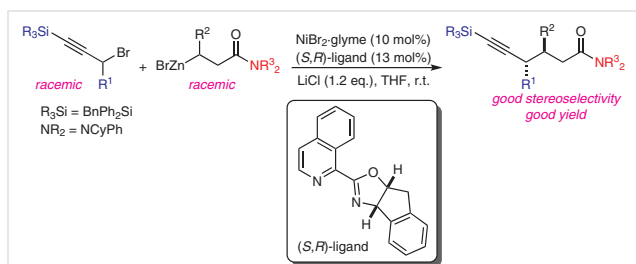
Scheme 18

Melzig and Knochel et al.²³ reported a novel and efficient Ni-catalyzed cross-coupling reaction of highly functionalized organozinc reagents with thiomethylated N-heterocycles that was achieved at room temperature, furnishing the expected heterocyclic products in 68–96% yield (Scheme 19). This protocol utilized various kinds of zinc reagents (alkyl, aryl, heteroaryl, and benzylic), and tolerates a wide range of functional groups.



Scheme 19

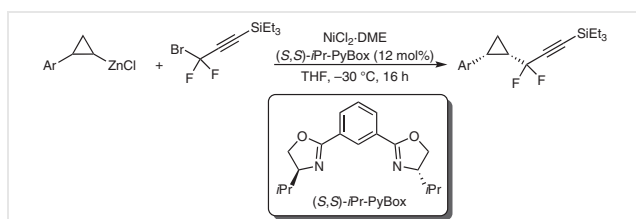
Huo et al.²⁴ reported a chiral nickel catalyst that coupled racemic electrophiles (propargylic halides) with racemic nucleophiles (β-zincated amides) to form carbon–carbon bonds in doubly stereoconvergent processes, affording a single stereoisomer of the product from two stereochemical mixtures of reactants (Scheme 20). Efforts to extend this protocol to alkyl–alkyl coupling were complicated by the prospect of forming two different three-dimensional



Scheme 20

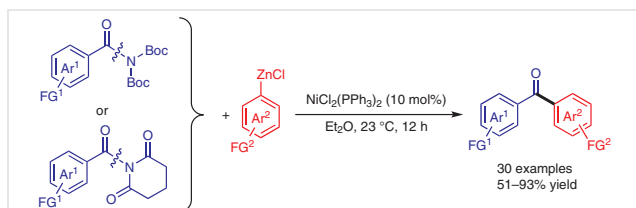
configurations at each carbon center, corresponding to four possible products.

Zhang et al.²⁵ reported an enantiodivergent strategy to stereoselectively access chiral cyclopropanes based on nickel-catalyzed enantioselective coupling of racemic *cis*-cyclopropyl-zinc reagents (Scheme 21). The reaction with alkyl electrophiles in particular, and with racemic secondary alkyl electrophiles, as well as the subsequent stereospecific relay coupling, provided a new strategy for asymmetric synthesis.



Scheme 21

Shi and Szostak^{21,26} developed a general cross coupling of primary amides enabled by the combined site-selective *N,N*-di-Boc activation and use of nickel catalysis through direct metal insertion into the N–C bond (Scheme 22). The reaction showed excellent functional group tolerance, providing rapid access to functionalized ketones from simple and readily available primary amides by selective N–C cleavage.

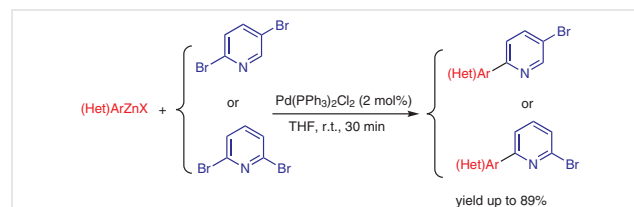


Scheme 22

2.3 Palladium-Catalyzed Cross Couplings Involving Organozinc Reagents

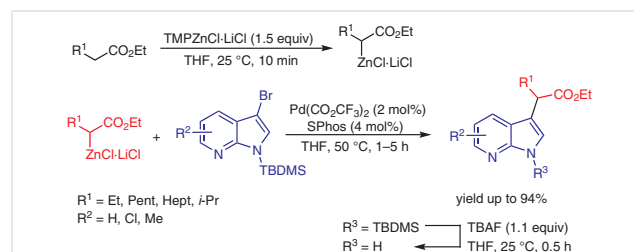
Jung and Kim²⁷ explored Pd(II)-catalyzed highly site-selective coupling reactions between (hetero)arylzincs and 2,5-dibromopyridine in THF solvent at room temperature

for 30 min, and the monocoupled compound was obtained as a major product. The 2,6-dibromopyridine was also demonstrated when employing the same equivalent of organozinc in this protocol under the same conditions (Scheme 23). It is worth noting that the mono-coupled products have the potential to be further transformed into highly substituted pyridine derivatives due to the presence of a bromo group on the pyridine ring.



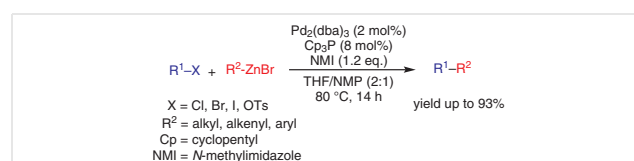
Scheme 23

Barl and co-workers²⁸ developed a process involving various Reformatsky reagents by the reaction of the corresponding carboxylic esters and TMPZnCl·LiCl; subsequently, the Reformatsky reagents reacted with TBDMS-protected 3-bromo-7-azaindoles and underwent the coupling reaction in the presence of Pd catalysis, furnishing a wide range of arylated ester derivatives. Finally, after elimination of the protecting group, the corresponding free 2-(7-azaindoly)carboxylic esters were obtained in satisfactory yields (Scheme 24).



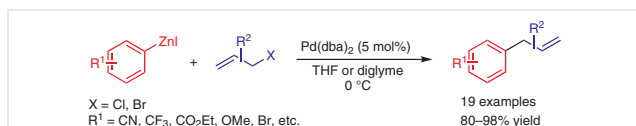
Scheme 24

Zhou and Fu²⁹ developed an efficient catalytic system (Pd₂(dba)₃/PCyp₃/NMI in THF at 80 °C) to achieve the Negishi cross coupling of a range of unactivated alkyl electrophiles with an array of alkyl-, alkenyl-, and arylzinc halides (Scheme 25). This process was compatible with a variety of functional groups, such as esters, amides, imides, nitriles, and heterocycles.



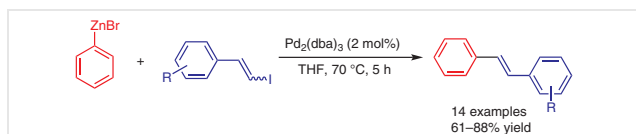
Scheme 25

Takagi and co-workers³⁰ developed an effective catalysis protocol using $\text{Pd}(\text{dba})_2$ to synthesize various functionalized allylbenzene derivatives by treating the ArZnX with allylic halides in THF or diglyme (Scheme 26). The $\text{Pd}(\text{dba})_2$ exhibited an outstandingly efficient catalytic effect for the Negishi cross coupling, and the reactions were mostly carried out at 0 °C for 5–30 min in the presence of 5 mol% catalyst.



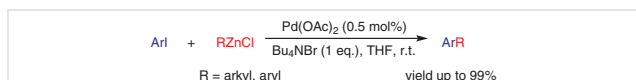
Scheme 26

Cook and co-workers³¹ developed a simple and general route for Pd-catalyzed Negishi cross coupling from arylvinyl iodides and arylzinc reagents to construct the $\text{C}(\text{sp}^2)\text{--C}(\text{sp}^2)$ bonds (Scheme 27). This system permitted efficient and selective preparation of *E*-stilbenes and their analogues in good yield, and it functioned effectively at low levels of catalyst loading without additional ligand. Furthermore, a wide range of functional groups were tolerated in this protocol.



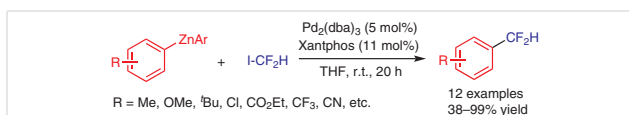
Scheme 27

Lei and co-workers³² developed an effective $\text{Pd}(\text{OAc})_2$ catalyzed Negishi coupling in the presence of Bu_4NBr between alkylzinc reagents or arylzinc reagents and aryl iodides at room temperature, giving the desired products in moderate to good yields (Scheme 28). This protocol has been successfully applied in $\text{Csp}^2\text{--Csp}^3$, $\text{Csp}^2\text{--Csp}^2$, and $\text{Csp}\text{--Csp}^2$ couplings.



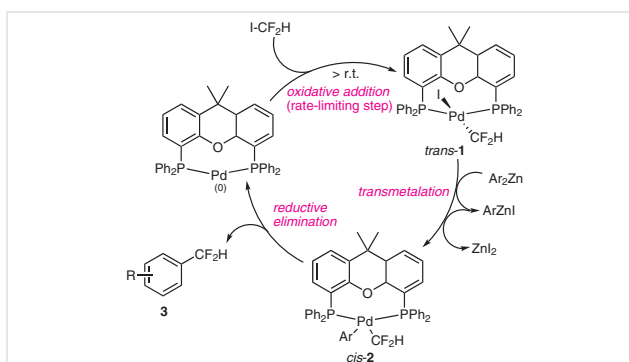
Scheme 28

Mikamia and co-workers³³ developed an efficient Negishi coupling starting from difluoroiodomethane with arylzincs in the presence of $[(\text{Xantphos})\text{Pd}^0]$ catalyst under mild reaction conditions, affording the corresponding difluoromethyl arenes (Scheme 29). This operationally simple methodology has proved to be efficient and environmentally friendly, giving the products in moderate to excellent yields.



Scheme 29

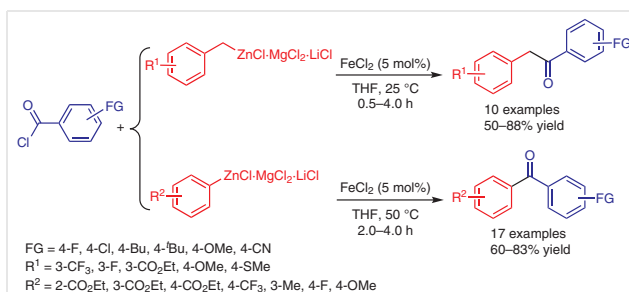
Mikamia also proposed a plausible reaction mechanism³³ based on their research (Scheme 30). Initially, oxidative addition of difluoroiodomethane to $[(\text{Xantphos})\text{Pd}^0]$ led to *trans*-1. Subsequently, a transmetalation with Ar_2Zn produced *cis*-2. The resultant ArZnI was also effective for transmetalation activated by TMEDA. Finally, reductive elimination afforded the desired coupling product 3. The structural change from *trans*-1 to *cis*-2 via transmetalation led to smooth reductive elimination.



Scheme 30 Possible mechanism of Pd-catalyzed Negishi coupling of difluoroiodomethane with arylzincs

2.4 Iron- and Copper-Catalyzed Cross Couplings Involving Organozinc Reagents

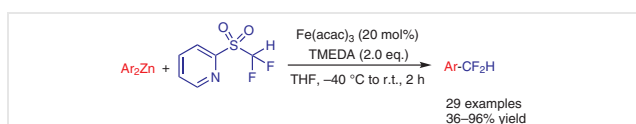
Iron salts are particularly attractive because of their cost-effective, environmentally friendly, and low-toxicity properties. Benischke and Knochel et al.³⁴ developed an efficient and practical iron-catalyzed cross-coupling reaction between various acid chlorides and functionalized benzylic zinc or arylzinc halides under mild reaction conditions (25–50 °C) in short time (0.5–4 h), which provided a variety of polyfunctionalized diaryl and aryl heteroaryl ketones in



Scheme 31

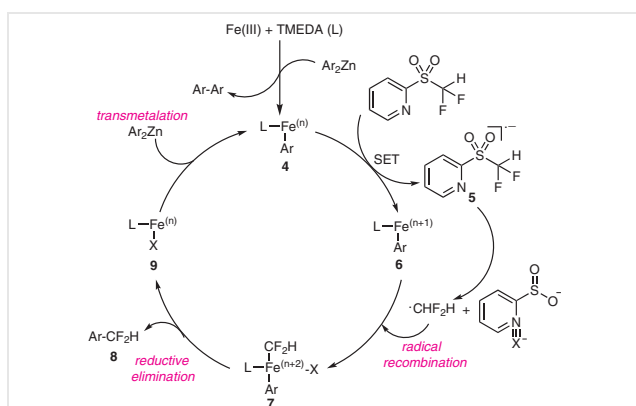
good yield (Scheme 31). The protocol enjoyed mild reaction conditions with broad substrate scope and good functional group tolerance.

Hu et al.³⁵ developed iron-catalyzed difluoromethylation between difluoromethyl 2-pyridyl sulfone and functional arylzincs and achieved selective C–S bond cleavage, creating facile access to the structurally diverse difluoromethylated arenes at low temperature (Scheme 32). This new approach employed a cost-effective iron catalyst and readily available, bench-stable fluoroalkyl sulfone reagent, giving new insights into the new reactivity of fluoroalkyl sulfones under transition-metal catalysis.



Scheme 32

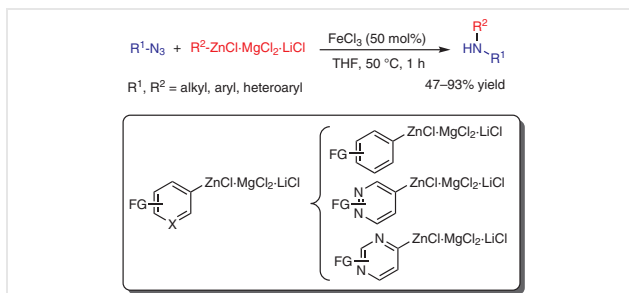
Based on the results obtained and on previous reports, the proposed reaction mechanism is also summarized in Scheme 33. The catalytic cycle commenced with the formation of a reduced iron species **4**, which was generated from the reduction of the Fe(acac)₃ precatalyst with an arylzinc reagent in the presence of TMEDA. Electron transfer between the catalytically active iron species **4** and 2-PySO₂CF₂H afforded the radical anion **5**, the fragmentation of which produced a difluoromethyl radical that underwent recombination with iron complex **6** to produce intermediate **7**. Subsequent reductive elimination delivered the desired product **8** and rendered the low-valent iron species **9**. The catalytically active iron species **4** was then regenerated through transmetalation between **9** and Ar₂Zn to close the catalytic cycle.



Scheme 33 Possible mechanism for iron-catalyzed difluoromethylation between difluoromethyl 2-pyridyl sulfone and functional arylzincs

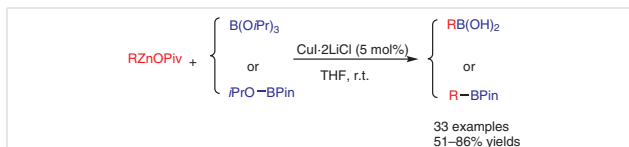
Grassl, Singer, and Knochel³⁶ reported the general electrophilic amination between highly functionalized alkyl, aryl and heterocyclic azides and various alkyl-, aryl- and heteroarylzinc halides in the presence of FeCl₃ (0.5 equiv) at 50 °C for 1 h, furnishing the corresponding secondary

amines in good yields (Scheme 34). This method could be extensively used in the synthesis of peptidic azides, providing the arylated substrates with full retention of configuration. Moreover, the iron salts could be used to trigger this amination effectively and efficiently.



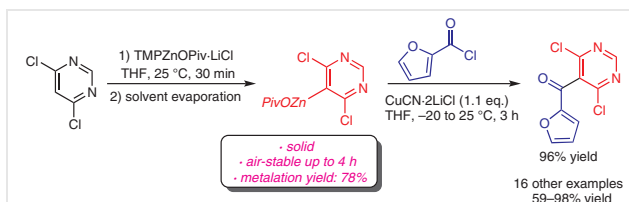
Scheme 34

Shen et al.³⁷ reported a practical and efficient copper-catalyzed protocol for constructing the C–B bond from trialkyl borates and air- and moisture-stable organozinc pivalates under mild conditions (THF as solvent at room temperature) (Scheme 35). This method avoided the generation of by-products, such as borane, and provided a more practical method for boronic ester synthesis in high yields.



Scheme 35

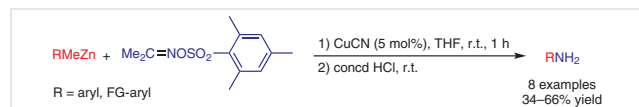
Stathakis, Manolikakes, and Knochel³⁸ developed a method for the preparation of a wide range of aryl and heteroaryl zinc pivalates bearing sensitive functionalities by selective metalation using TMPZnOPiv-LiCl; the obtained zinc reagent then smoothly underwent various cross coupling or acylations with a range of electrophiles (Scheme 36). It is worth mentioning that the new zinc reagents were easily processed to generate powders that exhibited excellent air stability.



Scheme 36

Kalkan et al.³⁹ reported an efficient selective amination reaction of mixed methylarylzincs with acetone O-(mesitylenesulfonyl)oxime in THF in the presence of CuCN at room

temperature and generated arylamines in moderate to good yields (Scheme 37). With the great range of potential applications, decent yields and low cost, this procedure provides an atom-economic alternative to existing methods of amination of diarylzincs for arylamine synthesis.



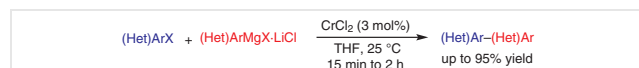
Scheme 37

3 Transition-Metal-Catalyzed Cross Couplings Involving Organomagnesium Reagents

Magnesium organometallic compounds have strongly influenced the development of organic synthesis in the last century. In recent years, the application of highly functionalized organomagnesium reagents has become an important method for the construction of complex organic target molecules. Furthermore, compared to organic zinc reagents, the organomagnesium boasts higher reactivity and therefore, achieves better yields under the reaction with electrophilic reagents.

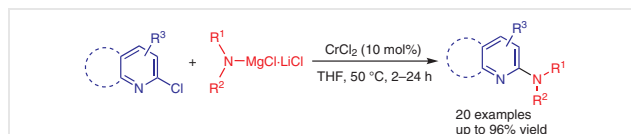
3.1 Chromium-Catalyzed Cross Couplings Involving Organomagnesium Reagents

Chromium salts have an acceptably low toxicity and are often used as an alternative metal catalyst that offers great potential in organic synthesis. Steib and Knochel et al.⁴⁰ developed a new, efficient and low toxic chromium(II) chloride-catalyzed cross-coupling reaction of various (hetero)-aryl magnesium reagents with N-heterocyclic halides, aromatic halogenated ketones or imines and alkenyl iodides at room temperature (Scheme 38). Ligand-free conditions, short reaction time, low toxicity and high reactivity chromium salt catalysts (only 3 mol%) are notable features of this protocol.



Scheme 38

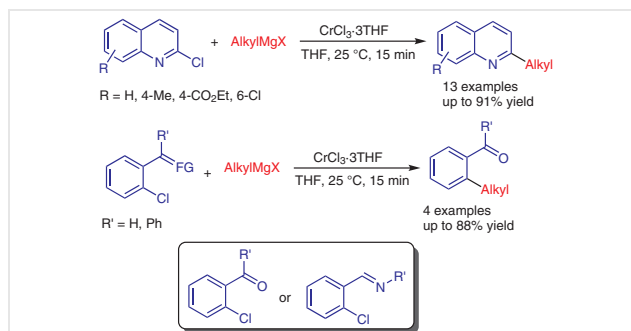
Steib and Knochel et al.⁴¹ also reported a ligand-free and selective chromium(II)-catalyzed amination of various aliphatic, allylic, benzylic, and saturated (hetero)cyclic magnesium amides with a range of N-heterocyclic mono- and dichlorides in the presence of lithium chloride as additive (Scheme 39). The reactions were performed at 50 °C in THF



Scheme 39

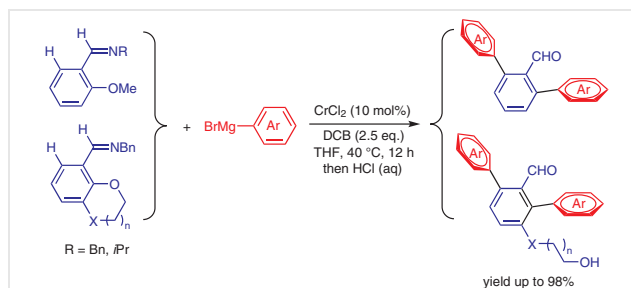
within short reaction time and led to the desired 2-aminated heterocycles in 56–96% yield.

Bellan and Knochel et al.⁴² reported an efficient method for chromium(III)-catalyzed C(sp²)-C(sp³) cross-coupling reactions of halo-quinolines and activated aryl chlorides with alkylmagnesium reagents in THF at 25 °C within 15 minutes, leading to the corresponding alkylated products (Scheme 40). This protocol employed tetrahydrofuran-soluble chromium(III) complex CrCl₃·3THF as catalyst, featured no ligand, and tolerated a number of functional groups.



Scheme 40

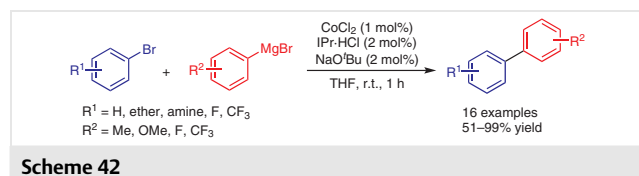
Zeng et al.⁴³ developed a cost-effective chromium-catalyzed one-pot method for a sequential difunctionalization of unactivated C–O and C–H bonds. This reaction was promoted by an inexpensive chromium(II) salt with an isopropyl or N-benzyl imino-substituted auxiliary, allowing for the sequential incorporation of two different or identical aromatic scaffolds into the benzaldehyde derivatives, and it offers a rare example of the sequential cleavage of unactivated C–O and C–H bonds for the formation of two aromatic C–C bonds in a simple one-pot operation (Scheme 41).



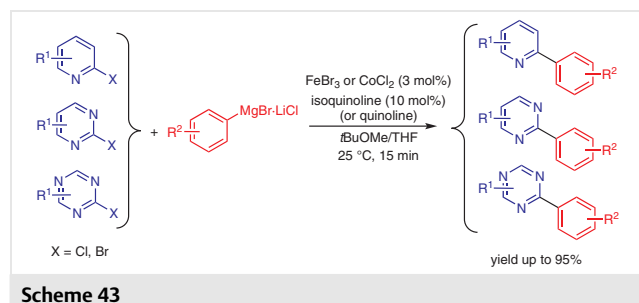
Scheme 41

3.2 Cobalt and Iron-Catalyzed Cross Couplings Involving Organomagnesium Reagents

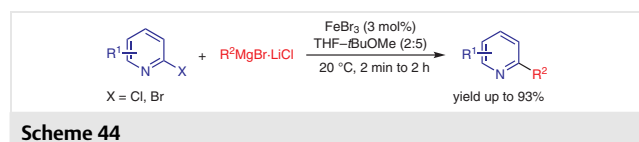
Wu and Duong⁴⁴ reported a highly active cobalt catalyst system comprised of $\text{CoCl}_2/\text{IPr}\cdot\text{HCl}/\text{NaOtBu}$ for cross-coupling reactions of aryl bromides and arylmagnesium reagents, that allowed a wide range of (hetero)biaryls to be prepared in 51–99% yields in THF at room temperature in one hour (Scheme 42). The advantages of this protocol included mild reaction conditions, short reaction time, good yield, and broad substrate range.



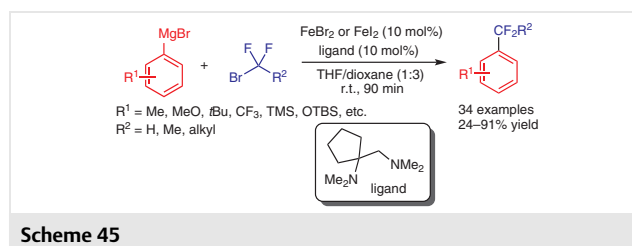
Kuzmina and Knochel et al.^{45,46} developed a novel and efficient catalytic system in which the isoquinoline (and quinoline) acted as a new efficient ligand for Fe- and Co-catalyzed cross couplings between N-heteroaryl halides and aryl magnesium reagents at room temperature for 15 minutes (Scheme 43). In this protocol, the quinoline and isoquinoline dramatically increased the catalytic efficiency of Fe and Co catalysts, and extended the reaction generality to fit with complex functional groups.



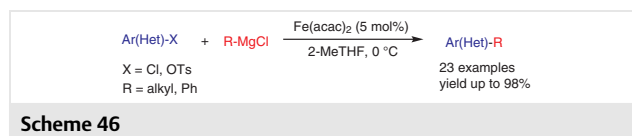
Kuzmina and Knochel et al.⁴⁷ also reported a new, simple and practical Fe-catalyzed $\text{sp}^2\text{-sp}^2$ cross coupling from various electron-rich or electron-poor arylmagnesium reagents and N-heterocyclic chlorides or bromides in mixed solvent (THF/*t*BuOMe) at mild temperature (Scheme 44). Mild, ligand-free reaction conditions, short reaction time, and tolerance of functional groups are advantages of this coupling reaction.



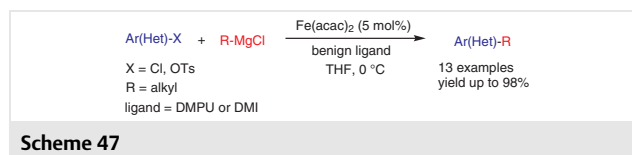
Zhang et al.⁴⁸ developed a novel and efficient diamine/iron catalytic system for the coupling reaction of difluoroalkyl bromides with arylmagnesium bromides under mild reaction conditions (THF/dioxane = 3:1, 25 °C for 90 min), and generated a number of difluoroalkylated products in good yields. The catalytic system used FeI_2 or FeBr_2 as the catalyst and modified *N,N,N',N'*-tetramethylethane-1,2-diamine (TMEDA) as a ligand, which dramatically improved the catalytic efficiency towards the cross coupling difluoroalkylation and suppressed the side reactions of de-fluorination (Scheme 45). This approach represented the first example of iron-catalyzed cross coupling between Grignard reagents and fluoroalkyl halides, and, notably, the vulnerable BrCF_2H was appropriate for this protocol.



Bisz and Szostak⁴⁹ developed an efficient iron-catalyzed cross coupling of aryl chlorides and tosylates with challenging organometallics possessing β -hydrogens using sustainable and eco-friendly 2-methyltetrahydrofuran (2-MeTHF) as solvent, which furnished the desired products in good to excellent yields (Scheme 46). This protocol featured excellent functional group tolerance under very mild, operationally convenient conditions.



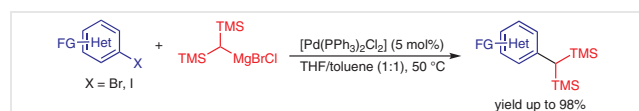
Bisz and Szostak⁵⁰ also developed efficient, sustainable ligands (DMI, DMPU) for iron-catalyzed $\text{C}(\text{sp}^2)\text{-C}(\text{sp}^3)$ coupling of aryl chlorides and tosylates with alkyl Grignard reagents (Scheme 47). This environmentally benign method accomplished traditionally difficult aryl-alkyl cross coupling with organometallics possessing β -hydrogens. The method was compatible with a variety of electrophilic functional handles.



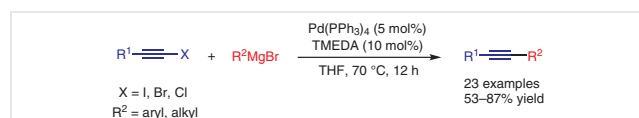
3.3 Pd, Ni and Mn-Catalyzed Cross Couplings Involving organomagnesium Reagents

Klatt and co-workers⁵¹ developed a simple procedure for the preparation of functionalized bis(trimethylsilyl)methyl (BTSM)-substituted heteroaryls from various heteroaryl halides and bis(trimethylsilyl)methyl magnesium bromide reagents catalyzed by palladium (Scheme 48). The reactions took place in mixed solvent (THF/toluene=1:1) at 50 °C. Furthermore, the BTSM group could then be converted into formyl, methyl, or styryl groups to give access to a variety of highly functionalized heteroaromatics.

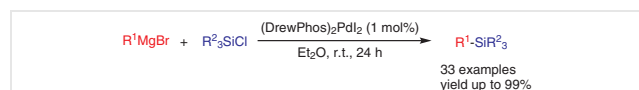
Li and co-workers⁵² developed an efficient and novel palladium-catalyzed Kumada cross-coupling reaction of Grignard reagents (aryl and alkyl Grignard reagents) with a wide range of alkynyl halides by using *N*¹,*N*¹,*N*²,*N*²-tetramethylethane-1,2-diamine (TMEDA) as ligand, providing the corresponding internal alkynes in moderate to good



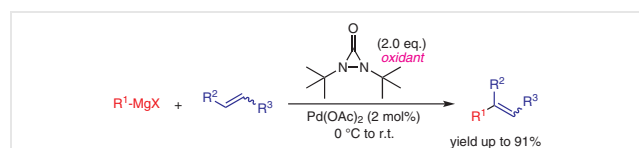
Scheme 48



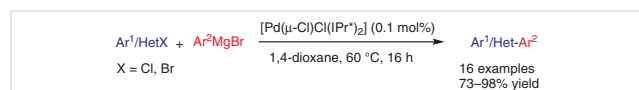
Scheme 49



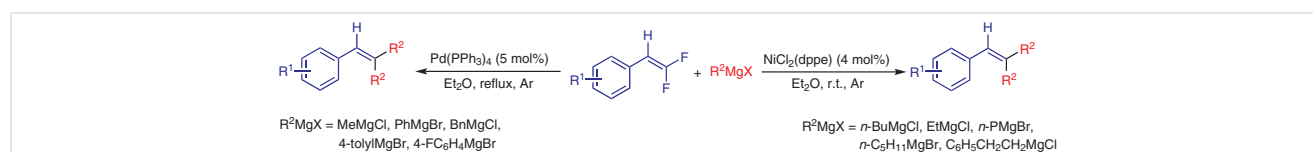
Scheme 50



Scheme 51



Scheme 52



Scheme 53

yields (Scheme 49). This novel protocol provided a mild and general access to internal alkynes with excellent functional group tolerance.

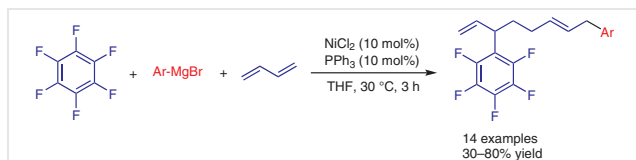
Watson and co-workers⁵³ developed a general palladium-catalyzed alkylation of chlorosilanes with primary and secondary alkylmagnesium halides under ligand-free conditions, which offered a vast range of alkyl or aryl silanes at room temperature (Scheme 50). This methodology overcame the high bond strength of the Si–Cl bond and was a rare example of a transition-metal-catalyzed process involving its activation.

Shi and co-workers⁵⁴ developed an efficient Pd-catalyzed oxidative Heck reaction starting from Grignard reagents with various terminal and internal olefins by using di-*tert*-butyldiaziridinone as the oxidant under mild conditions, giving the corresponding substituted olefins in 46–91% yields with high regioselectivity (Scheme 51). This protocol represented the first Pd-catalyzed oxidative Heck reaction with Grignard reagents.

Cazin and co-workers⁵⁵ developed a highly hindered pre-catalyst $[\text{Pd}(\mu\text{-Cl})\text{Cl}(\text{IPr}^*)]_2$ for the synthesis of tetra-*ortho*-substituted biaryls starting from Grignard reagent with aryl, naphthyl or heteroaryl halides (Scheme 52). This palladium complex exhibited very high catalytic activity in the cross-coupling reaction, and it could enable formation of highly hindered C–C biaryl junctions from Grignard reagent with very sterically demanding partners.

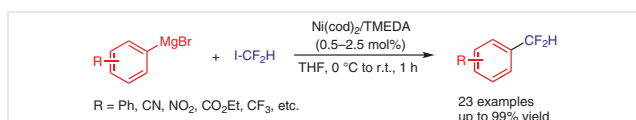
Cao and co-workers⁵⁶ developed two different palladium- and nickel-based catalytic systems for the Kumada–Tamao–Corriu cross-coupling reaction between *gem*-difluoroalkenes and Grignard reagents under simple and mild reaction conditions (room temperature or reflux in diethyl ether for 1–2 hours) without any additional ligands, leading to di-cross-coupled products in good to high yields (Scheme 53). In this methodology, the catalytic system involving $\text{Pd(PPh}_3)_4$ exhibited superior reactivity in the cross coupling of fluoroalkenes with Grignard reagents without β -hydrogen atoms, whereas $\text{NiCl}_2(\text{dppe})$ was exceptionally effective for the coupling reaction of fluoroalkenes with Grignard reagents bearing β -hydrogen atoms.

Kambe and co-workers⁵⁷ developed a regio- and stereo-selective multicomponent coupling reaction of perfluoroarenes, aryl Grignard reagents, and 1,3-butadiene in the presence of anionic Ni(II) complex, leading to the formation of various 1,6-octadiene derivatives containing two aryl groups (Scheme 54). This protocol provides a new strategy for the catalytic transformation of perfluoroarenes.



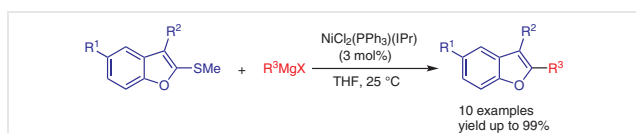
Scheme 54

Motohashi and Mikami⁵⁸ developed a novel nickel $[\text{Ni}(\text{cod})_2/\text{TMEDA}]$ -mediated cross-coupling reaction starting from the Grignard reagents with difluoroiodomethane under the mild reaction conditions within 1 h, providing the corresponding aromatic difluoromethylation products in excellent to moderate yields (Scheme 55). Wide substrate scope, short reaction time, low catalyst loading and mild reaction conditions were the features of this protocol.



Scheme 55

Osuka and co-workers⁵⁹ developed a highly efficient $\text{NiCl}_2(\text{PPh}_3)(\text{IPr})$ catalyzed alkylation of alkyl Grignard reagents with 2-methylsulfanylbenzofurans in THF at room temperature (Scheme 56). This method can be applied to the synthesis of many biomolecules, such as protein tyrosine phosphatase inhibitors [3-(4-biphenyl)-2-alkylbenzofurans].

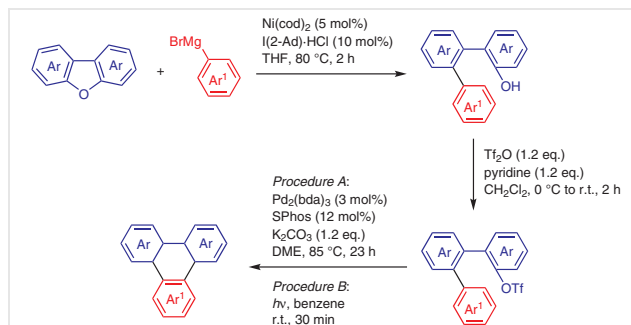


Scheme 56

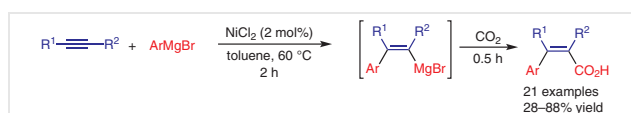
Osuka and co-workers⁶⁰ developed a new class of aromatic metamorphosis in which dibenzofurans were transformed into triphenylenes in three steps through sequential nickel-catalyzed C–O bond arylation, trifluoromethanesulfonylation of the resulting hydroxy moiety with TiF_4 , as well as palladium-catalyzed or photoinduced ring closure (Scheme 57). This protocol provided dorsally benzo-fused helicene in a satisfactory yield by employing π -extended dinaphthofuran as the substrate.

Wang and Xi⁶¹ developed a ligand-free nickel-catalyzed arylcarboxylation of alkyne with aryl Grignard reagents and carbon dioxide to afford trisubstituted β -arylacrylic acids with good regioselectivity under mild conditions in a one-pot process (Scheme 58). The resulting products could be further transformed into functionalized cycles.

A possible mechanism was proposed as shown in Scheme 59. Initially, $\text{Ni}(\text{II})$ is reduced by Grignard reagents to deliver $\text{Ni}(\text{0})$, which undergoes cycloaddition with alkyne

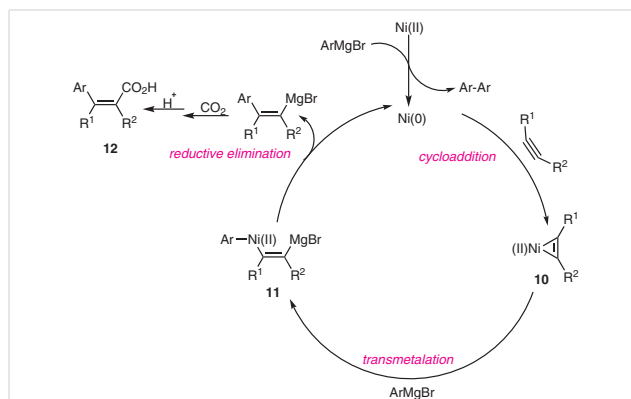


Scheme 57



Scheme 58

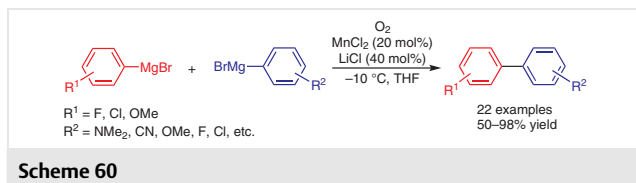
to gives $\text{Ni}(\text{II})$ complex **10**. Then the aryl Grignard reagent triggered transmetalation to afford new $\text{Ni}(\text{II})$ complex **11**. Finally, reductive elimination takes place to generate the arylmagnesium-substituted alkene, which gave acrylic acid **12** after addition with CO_2 and subsequent hydrolysis.



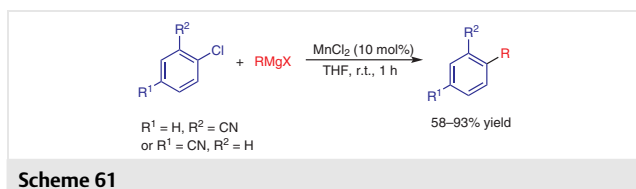
Scheme 59 Plausible mechanism for ligand-free nickel-catalyzed arylcarboxylation of alkyne with aryl Grignard reagents

Madsen and co-workers⁶² reported an improved protocol for the MnCl_2 -catalyzed aerobic cross coupling of two arylmagnesium bromides under an atmosphere of dioxygen (Scheme 60). The reaction was achieved with a 2:1 ratio between the Grignard reagents and 20% MnCl_2 . Under these conditions, very good yields of the heterocoupling product can be obtained wherein the limited Grignard reagents underwent very little homocoupling.

Madsen and co-workers⁶³ also reported MnCl_2 -catalyzed cross coupling of Grignard reagents and various aryl halides under mild reaction conditions for 1 h (Scheme 61). In this protocol, the reaction proceeded rapidly with good resulting yields when the aryl halide component was an



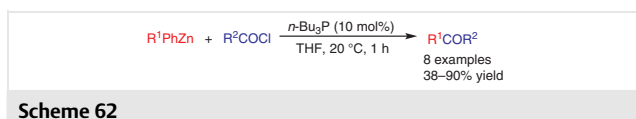
aryl chloride containing a cyano group in the *ortho* position or a cyano group in the *para* position. However, a range of other substituents either gave no conversion of the aryl halide or led to the formation of side products.



4 Transition-Metal-Free Cross Couplings Involving Zn and Mg Organometallic Reagents

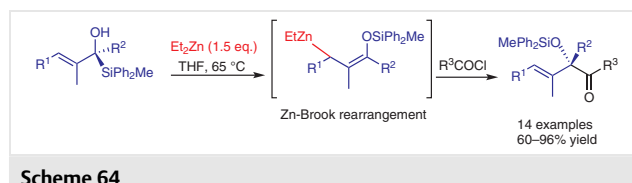
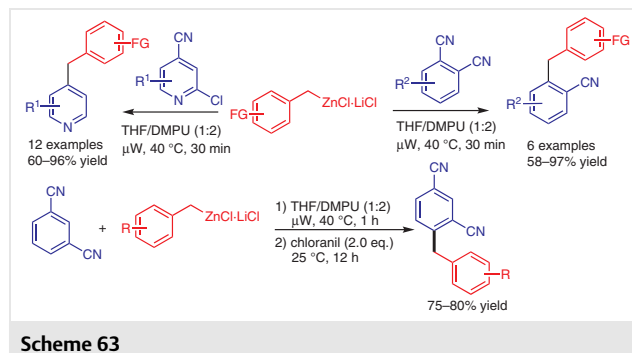
4.1 Transition-Metal-Free Cross Couplings Involving Organozinc Reagents

Erdik and Pekel⁶⁴ developed an efficient tri-*n*-butylphosphine-catalyzed acylation of aromatic acyl halides with *n*-alkyl phenylzincs in THF at 20 °C for one hour (Scheme 62). This method achieved the selective transfer of *n*-alkyl groups to *n*-alkyl aryl ketones in good yields, and it provided a simple and atom-economic organocatalyzed alternative to traditional transition-metal-catalyzed acylation of di-*n*-alkylzincs.



Quinio and Knochel et al.⁶⁵ developed a novel nucleophilic benzylation of polycyano-aromatics and 4-cyanopyridines using various readily available benzylic zinc chlorides in THF/DMPU under microwave irradiation (40 °C, 0.5–1.5 h), providing polyfunctional pyridines and cyano-substituted aromatics (Scheme 63). Furthermore, a novel oxidative nucleophilic substitution of a hydrogen using 1,3-dicyanobenzene with benzylic zinc reagents also took place in this procedure.

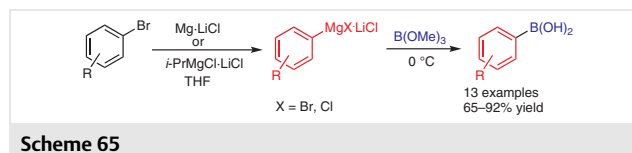
Marek and co-workers⁶⁶ developed enantiomerically enriched α -hydroxy allylsilanes, treated with Et₂Zn in THF at 65 °C that subsequently underwent Zn-Brook rearrangement to give configurationally stable allyl-zinc intermedi-



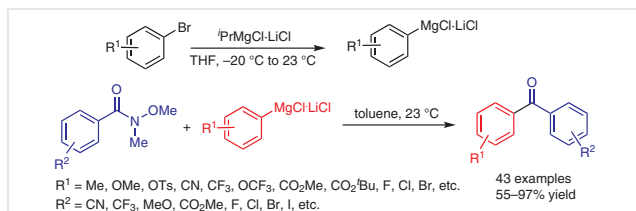
ates, which reacted with acyl chloride and methyl chloroformate with complete transfer of chirality (Scheme 64).

4.2 Transition-Metal-Free Cross Couplings Involving Organomagnesium Reagents

Colobert and co-workers⁶⁷ developed a general and convenient one-pot procedure for transforming inexpensive aryl bromides into the corresponding boronic acids under noncryogenic conditions. The electron-rich and sterically crowded Grignard reagents were very efficiently prepared by direct insertion of magnesium in the presence of LiCl or by Mg/Br exchange with *i*PrMgCl·LiCl. Subsequently, the Grignard reagents underwent electrophilic borylation reaction with B(OMe)₃, and were transformed into various aryl-boronic acids (Scheme 65). The methodology was very tolerant toward many functional groups, and various aryl-boronic acids were synthesized in a straightforward manner in excellent yields at 0 °C.

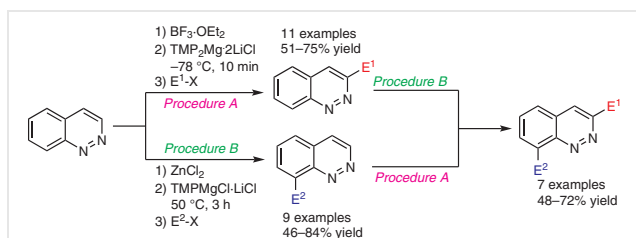


Li and Szostak⁶⁸ reported a simple and practical procedure for chemoselective preparation of functionalized biaryl ketones by arylation of functionalized Grignard reagents (prepared by Mg/halide exchange with *i*PrMgCl·LiCl in THF) with Weinreb amides (*N*-methoxy-*N*-methylamides) in toluene at 23 °C (Scheme 66). This transition-metal-free approach was demonstrated to exhibit excellent functional group tolerance.



Scheme 66

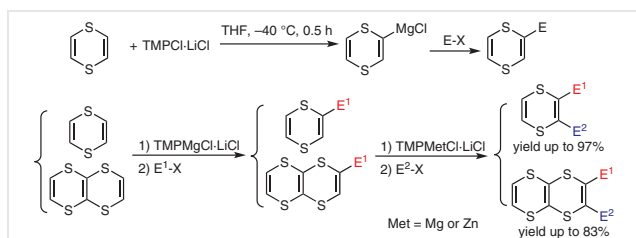
Klatt and Knochel et al.⁶⁹ developed a new general method for the regioselective metalation of the cinnoline scaffold by implementation of two complementary metalation procedures ($\text{BF}_3\cdot\text{OEt}_2$ and $\text{TMP}_2\text{Mg}\cdot 2\text{LiCl}$ or $\text{TMP}_2\text{Zn}\cdot 2\text{MgCl}_2\cdot 2\text{LiCl}$), allowing either a magnesiation at C(3) or a selective zincation at C(8). Various functionalizations by cross coupling reactions with electrophilic reagents were carried out successfully (Scheme 67). Additionally, a second selective metalation of prefunctionalized cinnoline derivatives could be performed, leading to highly functionalized 3,8-disubstituted cinnoline compounds.



Scheme 67

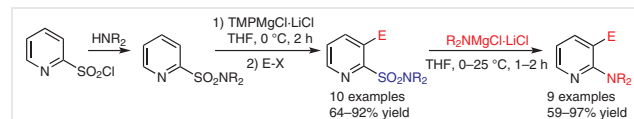
Castello-Mico and Knochel et al.⁷⁰ reported an efficient approach for the selective metalation of 1,4-dithiins by using $\text{TMPMgCl}\cdot\text{LiCl}$ and $\text{TMPZnCl}\cdot\text{LiCl}$, allowing facile magnesiation and zincation at the 1,4-dithiin scaffold, which produced polyfunctionalized 1,4-dithiins. A subsequent metalation of these *S*-heterocycles can also be achieved with the same TMP bases, leading to 2,3-disubstituted 1,4-dithiins (Scheme 68). This protocol demonstrated that $\text{TMPMgCl}\cdot\text{LiCl}$ and $\text{TMPZnCl}\cdot\text{LiCl}$ were excellent bases for facile metalation of the sensitive 1,4-dithiin scaffold.

Balkenhohl and Knochel et al.⁷¹ developed a new, mild transition-metal-free amination of related *N*-heterocycles and pyridine-2-sulfonyl chloride by using $\text{R}_2\text{NMgCl}\cdot\text{LiCl}$.



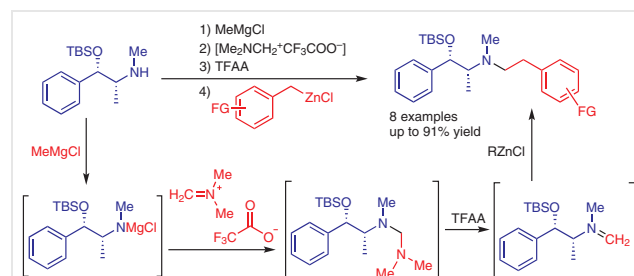
Scheme 68

Additionally, the pyridine-2-sulfonamides underwent a directed *ortho*-magnesiation reaction by using $\text{TMPMgCl}\cdot\text{LiCl}$. Reaction of the magnesium intermediates with various electrophiles and subsequent amination using magnesium amides led to a range of 2,3-functionalized pyridines in good yields (Scheme 69).



Scheme 69

Ellwart and Knochel et al.⁷² developed a one-pot homologative amination procedure for preparation of polyfunctional ephedrine derivatives starting from the two enantiomers of ephedrine using the Potier reagent, and the polyfunctional aryl- or benzylic organozinc halides (Scheme 70). This method has important values for the synthesis of molecules with pharmacological and physiological activity.

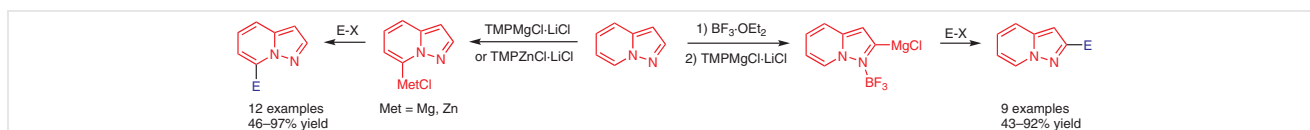


Scheme 70

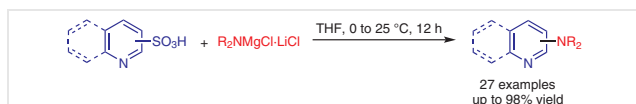
Balkenhohl and Knochel et al.⁷³ reported a regioselective functionalization of the pyrazolo[1,5-*a*]pyridine scaffold in the C2 and C7 position by using $\text{TMPMgCl}\cdot\text{LiCl}$ or $\text{TMPZnCl}\cdot\text{LiCl}$ (TMP = 2,2,6,6-tetramethylpiperidyl) in the presence or absence of the Lewis acid $\text{BF}_3\cdot\text{OEt}_2$. Various functionalized pyrazolo[1,5-*a*]pyridines bearing an ester group (NHBoc or ethyl group) were magnesiated and functionalized, leading to polysubstituted heterocycles (Scheme 71).

Balkenhohl, Valsamidou, and Knochel⁷⁴ developed a transition-metal-free mediated amination of 2-pyridine- and 8-quinolinesulfonic acids by using $\text{R}_2\text{NMgCl}\cdot\text{LiCl}$. Thus, several cyclic and acyclic amines were converted into the corresponding amides by using *i*-PrMgCl·LiCl, which reacted readily with sulfonic acids to generate 2-aminopyridines and 8-aminoquinolines (Scheme 72). Notably, this amination was readily scaled up to 50 mmol scale.

Balkenhohl and Knochel et al.⁷⁵ also reported an efficient amination of various phosphorodiamidate-substituted pyridines, quinolines, and quinoxaline with magnesium amides $\text{R}_2\text{NMgCl}\cdot\text{LiCl}$ proceeding at room temperature within eight hours. Several pharmaceutically active amines

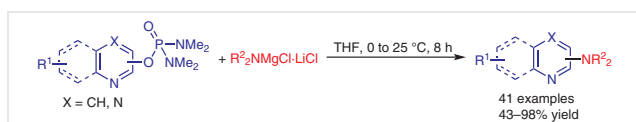


Scheme 71



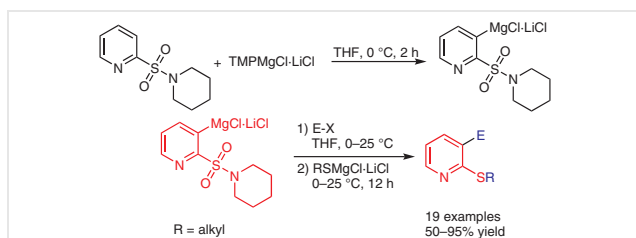
Scheme 72

were suitable substrates for this amination procedure (Scheme 73). Additionally, the C4 position of both pyridine and quinoline were readily aminated.



Scheme 73

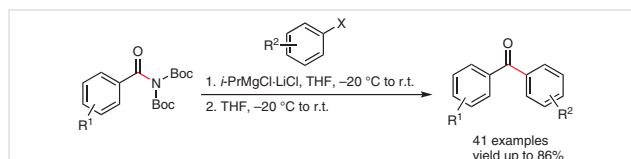
Heinz, Balkenhohl and Knochel⁷⁶ reported the thiolation of pyridine-2-sulfonamides by using magnesium thiolates. The *ortho*-functionalizations of these sulfonamides using TMPMgCl·LiCl followed by electrophilic quenching produced a range of 3-functionalized pyridine-2-sulfonamides, which were subsequently converted into the corresponding thioethers in good yield (Scheme 74). This protocol offered 19 examples and tolerated a wide range of functional groups.



Scheme 74

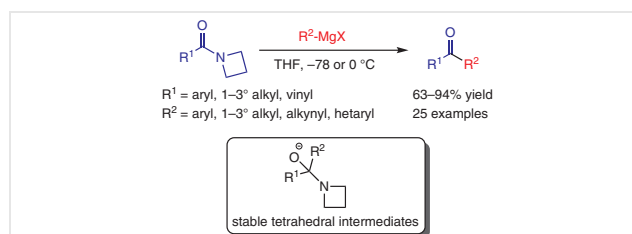
Li and Szostak^{4a} reported a transition-metal-free direct acylation of amides with functionalized Grignard reagents that proceeded by a highly chemoselective N–C cleavage (Scheme 75). This protocol provided rapid access to functionalized ketones by an unconventional amide bond acylation and featured exceedingly mild reaction conditions, simple operation, and wide functional group tolerance.

Szostak and co-workers⁷⁷ also reported a general and highly chemoselective method for the synthesis of ketones by the addition of organometallics to *N*-acylazetidines via stable tetrahedral intermediates (Scheme 76). The transfor-



Scheme 75

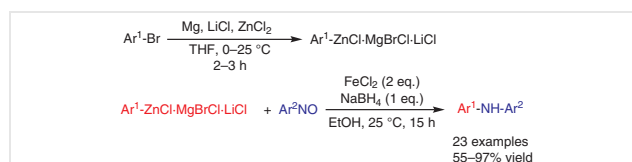
mation was characterized by its wide substrate scope and exquisite selectivity for the ketone products even when a large excess of nucleophilic reagent was used.



Scheme 76

5 Nucleophilic Additions Involving Zn and Mg Organometallic Reagents

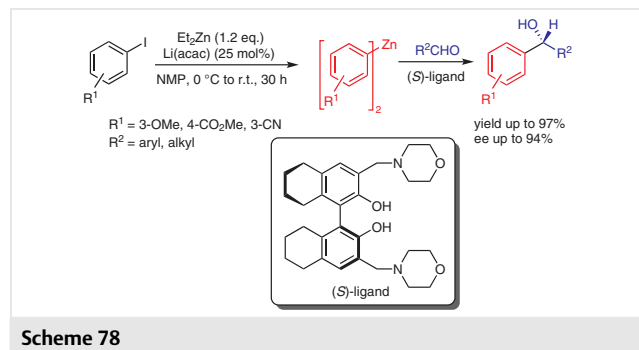
Dhayalan, Samann, and Knochel⁷⁸ developed an efficient and simple approach for the synthesis of polyfunctional secondary amines by the addition of functionalized aryl, heteroaryl or adamantyl zinc reagents to various nitrosoarenes. This reaction required reductive work-up with FeCl₂ and NaBH₄ in ethanol, and then produced the corresponding polyfunctional secondary amines in high yields (Scheme 77). Notably, both Mg and Li salts were necessary to achieve high yields for the synthesis of the corresponding functionalized secondary amines.



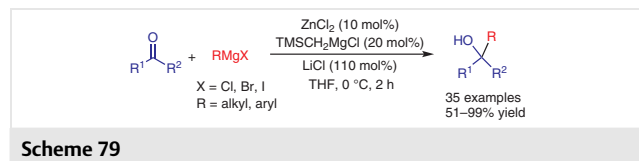
Scheme 77

Pu and co-workers⁷⁹ developed an easily available chiral ligand that could activate the nucleophilic reaction of arylzincs that were prepared in situ from aryl iodides and Et₂Zn. Subsequently, the nucleophilic addition reaction of aryl-

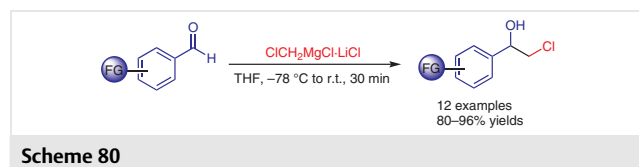
zincs with a variety of aldehydes took place, giving the desired products in high yields and high enantioselectivity (up to >99% ee) (Scheme 78). The mild reaction conditions and the good functional group tolerance made this catalytically asymmetric process synthetically useful.



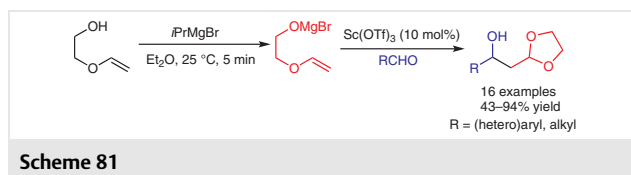
Ishihara et al.⁸⁰ developed highly efficient catalytic alkylation and arylation reactions starting from ketones, aldehydes, and aldimines with Grignard reagents by using LiCl as additive, which effectively minimizes side reactions (Scheme 79). This simple and robust protocol represents a breakthrough in the efficient alkylation of carbonyl compounds since a variety of commercially available Grignard reagents could be used.



Clososki et al.⁸¹ reported a highly chemoselective magnesium reagent ($\text{ClCH}_2\text{MgCl} \cdot \text{LiCl}$), a mixed lithium-magnesium carbenoid, to react with aromatic aldehydes bearing various functional groups, such as nitrile, ester, ketone, or amide groups, giving high yields of the corresponding functionalized aromatic chlorohydrins (Scheme 80). This protocol featured strong functional group tolerance, short reaction time, and excellent yield.

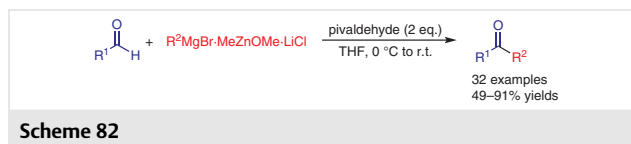


Quinio et al.⁸² developed an efficient procedure for the preparation of a broad range of functionalized protected aldol compounds via the addition of bromomagnesium reagent to various aldehydes in the presence of catalytic $\text{Sc}(\text{OTf})_3$ (Scheme 81). Access to enantioenriched alcohols was possible via a Swern oxidation-CBS reduction se-

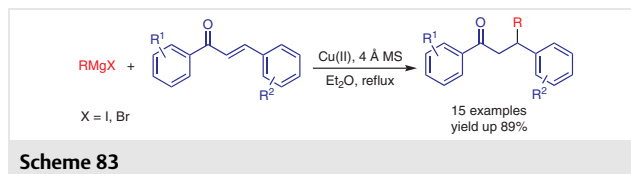


quence. It was shown that the dioxolane moiety could be deprotected under mild conditions.

Du et al.⁸³ developed a novel organozincation by using $\text{RMgX} \cdot \text{MeZnOMe} \cdot \text{LiCl}$, formed in situ via transmetalation of Grignard Reagent $\text{RMgBr} \cdot \text{LiCl}$ with MeZnOMe , which were excellent organometallic species in the nucleophilic addition/Oppenauer oxidation of aldehydes to generate aromatic ketones (Scheme 82). This protocol could be used to rapidly obtain structurally diverse aryl, heteroaryl, benzyl and alkyl ketones, with a broad substrate scope and excellent functional group tolerance.



Juhasz and Hell⁸⁴ developed a chemoselective addition of alkyl- and arylmagnesium halides to chalcones, catalyzed by copper(II) on a 4 Å molecular sieve support, and only the 1,4-addition products were obtained in high yields (Scheme 83). The catalyst in this protocol could be reused multiple times without significant loss of activity.

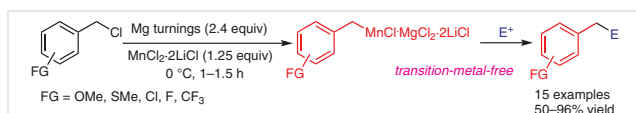


6 Cross Coupling Reactions or Nucleophilic Additions Involving Mn-, Al-, La-, Li-, Sm- and In-Organometallics

The Knochel group has researched extensively cross-coupling reactions starting from a variety of other organometallic reagents (not only organozinc or organomagnesium reagents). Thus, the results described herein are mainly related to Knochel's work.

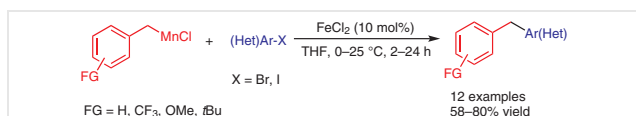
Functionalized organomanganese species have excellent chemoselectivity, low toxicity and good availability. Usually, their reactivity lies between those of organozinc and organomagnesium compounds. Quinio and Knochel et al.⁸⁵ developed a new and practical procedure for preparation of functionalized benzylic manganese reagents, which were smoothly prepared by the direct insertion of magnesium

into benzylic chlorides in the presence of $\text{MnCl}_2 \cdot 2\text{LiCl}$. Subsequently, the functionalized benzylic manganese reagents demonstrated their versatility in the presence of various electrophiles such as an allylic bromide, enone, aldehydes, and acid chlorides in the absence of any additional transition metal (Scheme 84). This reaction tolerates a range of functional groups and proceeds under mild conditions.



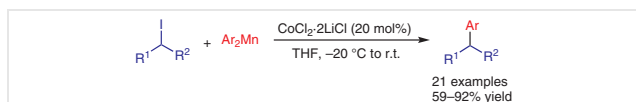
Scheme 84

Benischke and Knochel et al.⁸⁶ developed a convenient FeCl_2 (10 mol%) catalyzed cross-coupling reaction starting from various aryl and heteroaryl chlorides, bromides and iodides with benzylic manganese(II) chlorides, providing polyfunctionalized diaryl- and arylheteroarylmethane derivatives in good yield (Scheme 85). This method tolerates a range of functional groups, such as esters, nitriles or ketones and proceeds smoothly at room temperature within 2 to 24 hours.



Scheme 85

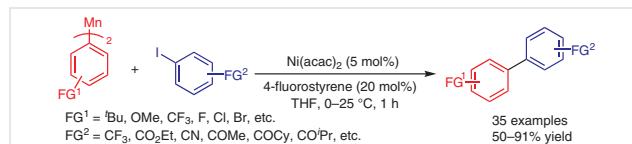
Hofmayer and Knochel et al.⁸⁷ developed a novel and efficient cobalt-catalyzed cross coupling of secondary alkyl iodides with polyfunctional diarylmanganese reagents by using the highly soluble cobalt salt $\text{CoCl}_2 \cdot 2\text{LiCl}$ in the absence of any additional ligand, which led to the cross-coupling products up to 92% yield (Scheme 86). In this protocol, high diastereoselectivities could be reached (dr up to 99:1), and no rearrangement of the secondary alkyl group occurred.



Scheme 86

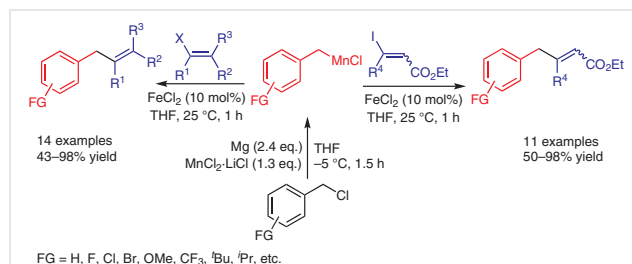
Benischke and Knochel et al.⁸⁸ developed an efficient catalytic system consisting of $\text{Ni}(\text{acac})_2$ (5 mol%) and 4-fluorostyrene (20 mol%) mediated cross coupling of a variety of aryl or heteroaryl halides with a range of functionalized organomanganese compounds, which afforded polyfunctionalized aryl- and arylheteroarylmethane derivatives in good yield (Scheme 87). The reaction scope was quite broad, including benzyl-, aryl- and alkylmanganese re-

agents as well as electrophiles bearing a sensitive ester or nitrile moiety, and had excellent functional group tolerance.



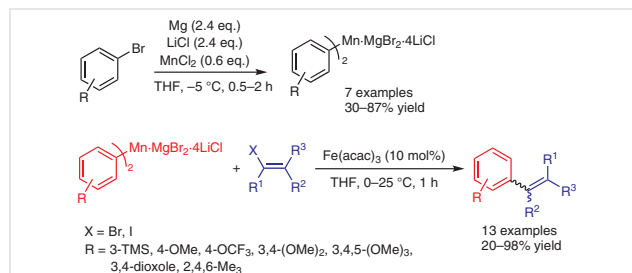
Scheme 87

Desaintjean and Knochel et al.⁸⁹ developed a simple approach for the preparation of various functionalized benzylic manganese chlorides from the corresponding benzylic chlorides by insertion of magnesium turnings in the presence of $\text{MnCl}_2 \cdot 2\text{LiCl}$ in THF at -5°C within two hours. Subsequently, these benzylic manganese reagents underwent smooth cross couplings with various functionalized alkenyl iodides, bromides, and triflates or iodoacrylates in the presence of 10 mol% FeCl_2 at 25°C for one to twelve hours (Scheme 88). A good functional group tolerance was a feature of this protocol.



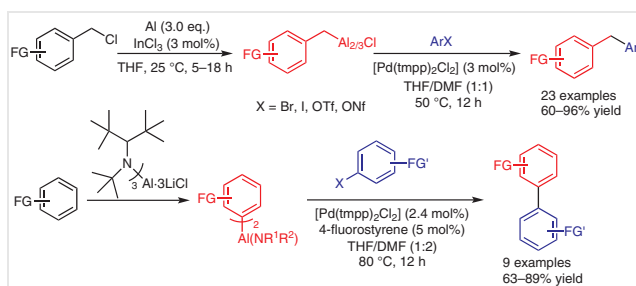
Scheme 88

Rousseau and Knochel et al.⁹⁰ developed a practical procedure for the preparation of various functionalized bis(aryl)manganese species from the corresponding aryl bromides by one-pot insertion of magnesium turnings in the presence of LiCl and in situ transmetalation with MnCl_2 in THF at -5°C within two hours. Subsequently, these bis(aryl)manganese reagents underwent smooth iron-catalyzed cross couplings using 10 mol% $\text{Fe}(\text{acac})_3$ with various functionalized alkenyl iodides and bromides (Scheme 89).



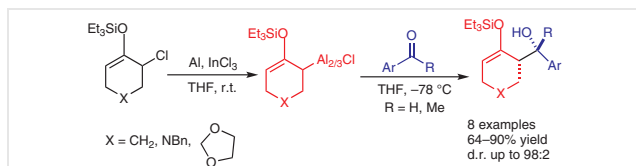
Scheme 89

Organoaluminums have become increasingly important in organic synthesis due to their unusually high chemoselective reactivity, and also because of the attractive price of aluminum and moderate toxicity. Groll and Knochel et al.⁹¹ developed an efficient and direct Pd-catalyzed cross coupling of functionalized organoaluminum reagents with organic halides. This methodology allowed a practical C–C bond formation starting from organoaluminum sesquihalides that were prepared from the corresponding benzyl chloride by an Al insertion, or using bis(organo)aluminum amides (Scheme 90). This protocol showed that aryl and heteroaryl iodides, bromides, and nonaflates, and, in special cases, chlorides and triflates, were good electrophiles for such cross couplings.



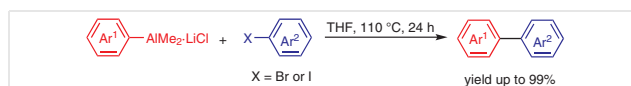
Scheme 90

Shen and Knochel et al.⁹² reported an efficient approach for preparation of aldol products bearing an α,β -quaternary center with high diastereoselectivities by using novel functionalized allylic aluminum reagents which tolerated a silyloxy substituent in the 2-position. The functionalized allylic aluminum reagents represented a new type of synthetic equivalent of metal enolates, and were prepared by conversion of chloro-substituted triethylsilyl enol ethers with aluminum powders in the presence of indium trichloride in THF at room temperature (Scheme 91). These allylic organoaluminum compounds underwent highly diastereoselective additions to aldehydes and methyl aryl ketones, giving aldol products with α,β -quaternary centers in good yields.



Scheme 91

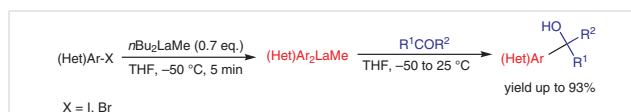
Minami and Knochel et al.⁹³ developed a novel transition-metal-free cross-coupling reaction between organic halides RX (R = aryl, alkenyl, alkynyl; X = I, Br, and Cl) and arylaluminum compounds ($\text{ArAlMe}_2\text{LiCl}$) under mild conditions, leading to a series of coupling products in good yield



Scheme 92

(Scheme 92). This reaction was efficient, selective, and applicable to a range of substrates.

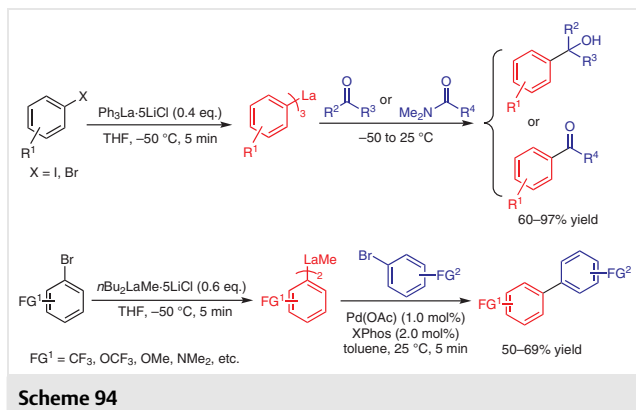
Organolanthanide(III) reagents have found many applications in organic synthesis, and they were good candidates for fine-tuning the character of the carbon–metal bond. The low electronegativity of lanthanum (1.1) comparable to that of lithium (0.98) confers to the carbon–lanthanum bond a high ionic character and therefore a high reactivity. Furthermore, the high valence of lanthanum(III) imparts this metal center with high oxophilicity and excellent Lewis acidity. Both of these properties have been extensively exploited for forming new carbon–carbon and carbon–heteroatom bonds. Benischke and Knochel et al.⁹⁴ developed a new functionalized alkylaluminum reagent by fast and convenient halogen–lanthanum exchange reactions starting from a variety of aryl or heteroaryl halides (X = Br, I) and $n\text{Bu}_2\text{LaMe}$, and a range of functionalized diaryl- and diheteroaryllanthanum derivatives were obtained at $-50\text{ }^\circ\text{C}$ within five minutes. Subsequent trapping reactions with selected electrophiles, such as ketones, aldehydes, or amides, proceeded smoothly at $-50\text{ }^\circ\text{C}$ in THF, affording polyfunctionalized alcohols and carbonyl derivatives (Scheme 93).



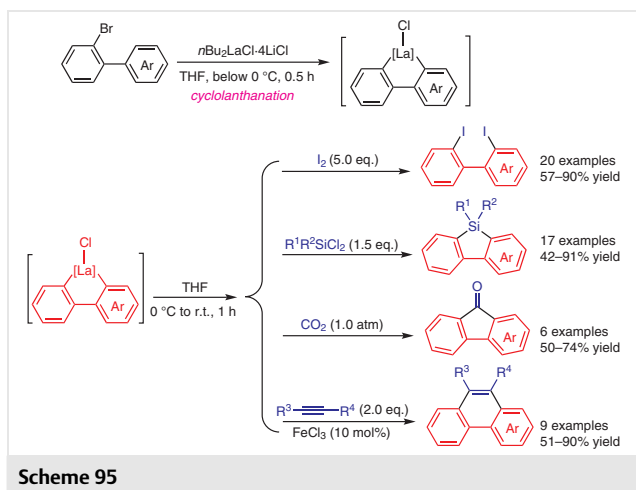
Scheme 93

Benischke and Knochel et al.⁹⁵ developed a novel method for the preparation of various triaryl- and triheteroaryl-lanthanum derivatives by halogen–lanthanum exchange reactions from various aryl halides (X = Br, I) and $\text{Ph}_3\text{La}\cdot 5\text{LiCl}$ or related ($m\text{-xylyl}$)₃La $\cdot 5\text{LiCl}$ in THF at $-50\text{ }^\circ\text{C}$ within five minutes. These new lanthanum organometallic reagents are compatible with isoquinolines and some functional groups such as a nitrile or an ester. Subsequently, the reactivity of the resulting lanthanum compounds with electrophiles, such as ketones, aldehydes, N,N -dimethylamides, and primary alkyl halides were investigated. Moreover, coupling occurred starting from the aryl bromides with the lanthanum compounds catalyzed by Pd (Scheme 94).

Wei and Knochel et al.⁹⁶ developed a novel and efficient protocol in which various aryl- and heteroaryl-substituted 2-bromobiaryls underwent smooth cyclolanthanation reaction below $0\text{ }^\circ\text{C}$ within 30 minutes with $n\text{Bu}_2\text{LaCl}\cdot 4\text{LiCl}$, and were converted into cyclometalated lanthanum intermediates. The resulting cyclometalated lanthanum reagents

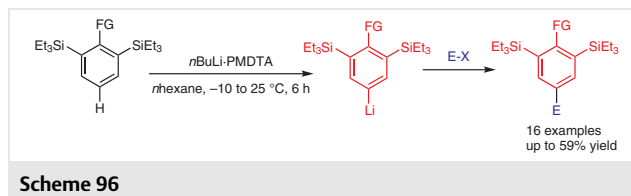


were key intermediates for the facile preparation of functionalized 2,2'-diiodobiaryls, silafluorenes, fluoren-9-ones, phenanthrenes and their related heterocyclic analogues (Scheme 95).

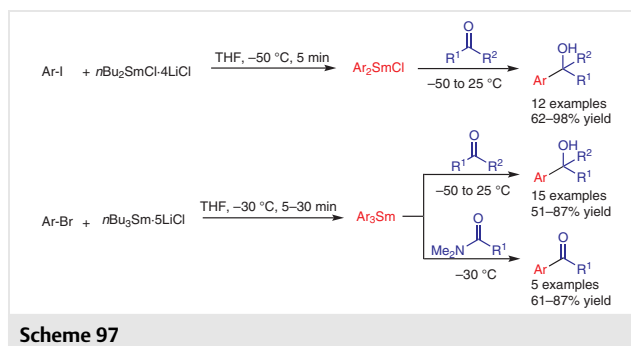


Organolithium compounds possess a highly polarized carbon–lithium bond, which ensures a high reactivity towards various electrophilic reagents, and they facilitate the retrosynthesis of complex organic molecules considerably. Bellan and Knochel⁹⁷ developed a remote and regioselective lithiation of functionalized arenes bearing two bulky triethylsilyl substituents at the 4-position (*para*-position) by using a mixture of *n*BuLi and PMDTA (*N,N,N',N'',N'''*-pentamethyldiethylenetriamine). The resulting aryllithium species, bearing sensitive functional groups such as amides or carbamates, reacted with a range of electrophiles leading to tri- and tetrasubstituted remotely functionalized products (Scheme 96). This lithiation could efficiently hamper *ortho*- and *meta*-lithiations.

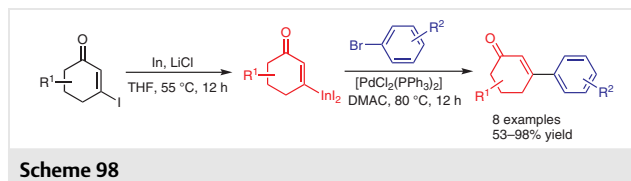
The organosamarium reagents are very useful organometallic intermediates with a better functional group tolerance (such as ester or pyridyl ring) than the corresponding aryllanthanum reagents. Anthore-Dalio and Knochel et



al.⁹⁸ developed a new I/Sm- and Br/Sm-exchange reaction in which various aryl and heteroaryl iodides and bromides were converted into the corresponding organosamarium reagents via halogen–samarium exchanges using *n*Bu₂SmCl and *n*Bu₃Sm, respectively. The resulting organosamarium reagents were efficiently quenched with aldehydes, ketones, and imines. They also underwent acylations when treated with *N,N*-dimethylamides, leading to ketones (Scheme 97). Remarkably, in contrast to the previously reported preparation of aryllanthanum species,⁹⁵ samarium reagents were compatible with more functional groups.



Organotin reagents are environmentally benign and display a low reactivity with both air and moisture. Moreover, they have shown good compatibility with water and several other functionalities. Shen and Knochel⁹⁹ developed a novel approach for the preparation of highly functionalized cyclo-alkenylindium(III) derivatives that could be directly inserted into cycloalkenyl iodides in the presence of LiCl in THF. In addition, the cyclo-alkenylindium with various aryl halides underwent cross coupling catalyzed by palladium, providing a series of coupling products in good yields (Scheme 98).



7 Conclusion

We have summarized the recent progress in cross-coupling reactions and nucleophilic additions involving

functionalized organometallic reagents. Although under transition-metal-catalyst or metal-free conditions, the cross-coupling reaction of organometallic and various electrophilic reagents has made great progress, the exploration of reaction pathways, methods to reduce leaching of transition metals during the cycle, and the use of environmentally friendly metal catalysts (such as Fe, Cu, Ni, etc.) still requires further investigation. On the other hand, the investigation of more efficient ligands and transition-metal catalysts and of more suitable reaction systems that could allow the reaction to proceed under mild conditions (open air in short time) are still challenges for organometallic reagents. In the future, the research hotspot on cross-coupling reactions involving organometallic reagents lies in the design and development of highly efficient catalysts that can be recycled multiple times, flow reactions, and the use of green solvents such as ionic liquids or supercritical fluids as the reaction medium to replace organic solvents.

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